MEASURABLE DENSITY BOUNDS HOLD FOR ALL COLOURINGS OF THE PLANE WITH ALGEBRAIC FORBIDDEN DISTANCES

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Abstract. OpenAI recently proved that every colouring of the plane with five colours gives the same colour to two points at distance 1. The proof transfers an arbitrary proper colouring of the plane to a *weak measurable* one, and then excludes weak measurable 5-colourings. We observe that the transfer works, with the same proof, for every set of algebraic forbidden distances and for fractional colourings. Upper bounds for the density of measurable sets avoiding given distances therefore become lower bounds for arbitrary colourings; all these lower bounds rest on OpenAI's rigidity theorem, which OpenAI formalized in Lean. Consequences: the fractional chromatic number of the plane equals $1/m_1(\mathbb{R}^2)$, where $m_1(\mathbb{R}^2)$ is the largest density of a measurable planar set with no two points at distance 1, so it lies between 4.0485 and 4.36; and the least independence ratio of an n-point unit-distance graph tends to $m_1(\mathbb{R}^2)$, as conjectured by Dúcz and Varga. If the distances 1 and 2 are both forbidden, at least 7 colours are needed, improving the bound 6 of Exoo and Ismailescu; the same holds for $\{1, \sqrt{3}\}$, and at least 8, 9 and 10 colours are needed for $\{1, 11/6\}$, $\{1, 40\}$ and $\{1, 100\}$. The distances 1, 100 and 10^4 force 27 colours, and 1, 10, 100 and 1000 force 26. For suitable sets of k integer distances more than 4.0485^k colours are needed, in the plane and in space, so the chromatic number with k forbidden distances grows exponentially in k, whereas Erdős expected polynomial growth. Davies has just proved the analogous statements in dimension at least 4 by a different method. The density bounds we use are linear-programming bounds of Oliveira Filho and Vallentin, which we certify in interval arithmetic, and theorems of Bukh and of Ambrus, Csiszárik, Matolcsi, Varga and Zsámboki.

1. Results

For a set D of positive real numbers, $\chi(\mathbb{R}^2, D)$ is the least number of colours needed to colour the plane so that no two points whose distance lies in D get the same colour. The Hadwiger–Nelson problem is the case $D = \{1\}$; OpenAI [18] proved $\chi(\mathbb{R}^2) \geq 6$, and $\chi(\mathbb{R}^2) \leq 7$ is Isbell's hexagonal colouring [21]. For two distances $\{1, d\}$ the best known lower bound was 6. Exoo and Ismailescu [11] proved $\chi(\mathbb{R}^2, \{1, 2\}) \geq 6$, Parts [19] found smaller graphs for it, and De Neve, Vanden Kerchove, Colle, Tavernier and Pickavet [8] gave 6-chromatic graphs for $d = \sqrt{3}$ and $d = (\sqrt{6} + \sqrt{2})/2$, two cases claimed earlier in the Polymath16 project [20]; Chybowska-Sokół, Junosza-Szaniawski and Węsek [3] record as open whether some pair $\{1, d\}$ forces 7 colours.

OpenAI's proof has two steps. Their Theorem 1.3 (the transfer) says that the plane has a proper k-colouring if and only if it has a *weak measurable* k-colouring: a measurable colouring in which the pairs of points at distance 1 with the same colour have measure zero. Their Theorem 1.4 excludes weak measurable 5-colourings. The transfer rests on a rigidity theorem for invariant measures on the character group of the algebraic plane [18, Thm. 2.3]. Its proof uses neither that the forbidden distance is 1 nor that a colouring uses finitely many colours rather than being fractional. We record the two generalisations (Theorems 8 and 9) and their consequences.

Write $m_D(\mathbb{R}^2)$ for the supremum of the upper densities of measurable sets $A \subseteq \mathbb{R}^2$ with no two points at a distance in D, and $m_1(\mathbb{R}^2) = m_{\{1\}}(\mathbb{R}^2)$. Croft's construction gives $m_1(\mathbb{R}^2) \geq 0.229364$ [5], and Ambrus, Csiszárik, Matolcsi, Varga and Zsámboki [1] proved $m_1(\mathbb{R}^2) \leq 0.2470$, confirming

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These results were obtained with the assistance of an AI system (Claude, by Anthropic), which found the observation, wrote the programs, ran the checks and helped to draft this note. Three separate AI agents refereed the note with their own programs. The hard input is OpenAI's transfer theorem [18]. No mathematician has checked the note yet. The author is not a professional mathematician and submits it for independent verification.

a conjecture of Erdős that $m_1(\mathbb{R}^2) < 1/4$. For the fractional chromatic number we use two versions, defined in Section 2: the finitary one $\chi_{\mathrm{f},0}(\mathbb{R}^2, D)$, the supremum over finite graphs drawn in the plane with edge lengths in D, and the fractional chromatic number $\chi_{\mathrm{f}}(\mathbb{R}^2, D)$ of the whole infinite graph.

Theorem 1. *Let D be a finite set of positive real algebraic numbers. Then*

$$\chi(\mathbb{R}^2, D) \;\geq\; \chi_{\mathrm{f},0}(\mathbb{R}^2, D) \;=\; \chi_{\mathrm{f}}(\mathbb{R}^2, D) \;=\; \frac{1}{m_D(\mathbb{R}^2)}.$$

The lower bound $\chi_{\mathrm{f},0}(\mathbb{R}^2, D) \geq 1/m_D(\mathbb{R}^2)$, and with it every lower bound below, rests on OpenAI's rigidity theorem [18, Thm. 2.3], which OpenAI formalized in Lean; the upper bound $\chi_{\mathrm{f}}(\mathbb{R}^2, D) \leq 1/m_D(\mathbb{R}^2)$ is elementary.

One distance.

Corollary 2. $\chi_{\mathrm{f},0}(\mathbb{R}^2) = \chi_{\mathrm{f}}(\mathbb{R}^2) = 1/m_1(\mathbb{R}^2)$, *and* $4.0485 < 1/0.2470 \leq \chi_{\mathrm{f}}(\mathbb{R}^2) \leq 1/0.229364 < 4.36$.

The constants in [1, §8] give $m_1(\mathbb{R}^2) < 0.2469975$, hence $\chi_{\mathrm{f}}(\mathbb{R}^2) > 4.0486$. Matolcsi, Ruzsa, Varga and Zsámboki [15] proved $\chi_{\mathrm{f}}(\mathbb{R}^2) \geq 4$ and conjectured that $\chi_{\mathrm{f},0}(\mathbb{R}^2) = 4$ and that $\chi_{\mathrm{f}}(G) < 4$ for every finite unit-distance graph G [15, Conj. 1]; the value 4 had also been conjectured in [1, Remark after Cor. 2]. Dúcz and Varga [10] disproved this conjecture: they found a unit-distance graph G_{29} whose geometric fractional chromatic number exceeds 4.0007, which gives a finite unit-distance graph with independence ratio below $1/4$ and $\chi_{\mathrm{f}}(\mathbb{R}^2) > 4.0007$. Corollary 2 gives a second disproof and determines $\chi_{\mathrm{f}}(\mathbb{R}^2)$ in terms of $m_1(\mathbb{R}^2)$. It also answers the question of [15] whether $\chi_{\mathrm{f},0}(\mathbb{R}^2) = \chi_{\mathrm{f}}(\mathbb{R}^2)$, and with [15, Thm. 2] it shows that the finitary independence ratio of the plane equals the measurable one, which [15] remark would fail if their conjecture held. Let $f(n)$ be the least independence number of the unit-distance graph of a set of n points in the plane, the function of Erdős's question [14] answered by Dúcz and Varga.

Corollary 3. $f(n) \geq m_1(\mathbb{R}^2)\, n$ *for every n, and* $f(n)/n \to m_1(\mathbb{R}^2)$ *as* $n \to \infty$. *In particular* $f(n) < 0.2470\, n$ *for all large n.*

The first statement is the classical averaging bound $m_1(\mathbb{R}^2) \leq \alpha_1(\mathbb{R}^2)$ [1, Remark after Cor. 2], [15, (14)], [10, Claim 1]; the new content is the limit, that is, $\alpha_1(\mathbb{R}^2) = m_1(\mathbb{R}^2)$. Dúcz and Varga conjectured that $f(n)/n = m_1(\mathbb{R}^2) + o(1)$, and in particular that $m_1(\mathbb{R}^2)$ is the density $0.2293647\ldots$ of Croft's construction [10, Conj. 1]. Corollary 3 proves the first statement; the second remains open.

Two distances.

Corollary 4.

$$\chi(\mathbb{R}^2, \{1,2\}) \geq 7, \qquad \chi(\mathbb{R}^2, \{1,\sqrt{3}\}) \geq 7, \qquad \chi(\mathbb{R}^2, \{1,(\sqrt{6}+\sqrt{2})/2\}) \geq 7,$$
$$\chi(\mathbb{R}^2, \{1,11/6\}) \geq 8, \qquad \chi(\mathbb{R}^2, \{1,40\}) \geq 9, \qquad \chi(\mathbb{R}^2, \{1,100\}) \geq 10.$$

Moreover $\chi(\mathbb{R}^2, \{1,d\}) \geq 17$ *for every sufficiently large real algebraic number d.*

The finite bounds come from linear programs certified in Section 5, and the last statement from Bukh's theorem [2]; products of hexagonal colourings give $\chi(\mathbb{R}^2, \{1,d\}) \leq 49$. For *measurable* colourings, bounds of this kind were known: Oliveira Filho and Vallentin [17, §4] report, from their linear program, the bound $0.141577 < 1/7$ for the density of a measurable set avoiding the distances 1 and $j_{1,2}/j_{1,1} \approx 1.8309$, a ratio of zeros of J_1, and, combining Bukh's theorem with their bound for $m_1(\mathbb{R}^2)$, they show $\inf_{d_1,d_2} m_{\{d_1,d_2\}}(\mathbb{R}^2) \leq m_1(\mathbb{R}^2)^2 \leq 0.072046$; so pairs of distances whose density is close to $m_1(\mathbb{R}^2)^2$ need 14 colours for measurable colourings (17 with the bound of [1]). What is new here is that such bounds hold for arbitrary colourings; the bounds for $\{1,2\}$ and $\{1,\sqrt{3}\}$, which use clique inequalities, also seem to be new for measurable colourings.

Many distances.

Corollary 5. *For every $k \geq 1$ there is a set D of k positive integers with $\chi(\mathbb{R}^3, D) \geq \chi(\mathbb{R}^2, D) \geq m_1(\mathbb{R}^2)^{-k} > 4.0485^k$. Moreover*

$$\sup\{\chi_{\mathrm{f}}(\mathbb{R}^2, D) : D \subseteq F \cap (0,\infty),\ |D| = k\} = m_1(\mathbb{R}^2)^{-k},$$

where F is the field of real algebraic numbers.

Erdős [13] defined $L_2(r)$ as the largest chromatic number of the plane with r forbidden distances and asked: "Does $L_2(r)$ increase polynomially or exponentially? I expect that $L_2(r) < r^c$ for some absolute constant c." The question is Problem #706 on Bloom's list [12], where it is stated for finite point sets, which gives the same $L_2(r)$ by the de Bruijn–Erdős theorem; Croft, Falconer and Guy ask whether the growth is exponential [4, Problem G11] (as quoted by Bukh [2]), and the best lower bound in the plane was of order $k\sqrt{\log k}$ (see [2, 16]). By Corollary 5 and the product colouring,

$$4.0485^r < L_2(r) \leq 7^r,$$

so $L_2(r)$ grows exponentially, contrary to Erdős's expectation. Bukh [2] proved exponential growth for measurable colourings and remarked that it is "very likely that the usual chromatic number does not share this behavior"; Corollary 5 shows that it does, for integer distances. Davies [6] has just proved exponential growth for arbitrary colourings of $\mathbb{R}^d$, $d \geq 4$, with a different transfer (see the Remarks); Corollary 5 gives it for $d = 2$ and $d = 3$. Like all our lower bounds, it rests on [18, Thm. 2.3]. Bukh also conjectured that $\chi(\mathbb{R}^d, D)$ is bounded independently of D when the elements of D are algebraically independent [2]; Theorem 8 says nothing about such D.

The same linear programs give explicit sets (Table 2):

Corollary 6. $\chi(\mathbb{R}^2, \{1, 40, 1600\}) \geq 22$, $\chi(\mathbb{R}^2, \{1, 10, 100, 1000\}) \geq 26$ *and* $\chi(\mathbb{R}^2, \{1, 100, 10^4\}) \geq 27$.

These numbers are far below $m_1(\mathbb{R}^2)^{-k}$, which is approached only by very spread distances, but the sets are explicit and the bounds are certified.

None of these results comes with an explicit finite graph: as in [18], the finite graphs exist by compactness.

2. Notation

Identify the plane with $\mathbb{C}$. Let $F = \overline{\mathbb{Q}} \cap \mathbb{R}$, let $E = F(i)$ be the plane of algebraic points and $K = \{u \in E : |u| = 1\}$ its group of rotations, as in [18, Def. 2.1]. For $D \subseteq (0,\infty)$ and $X \subseteq \mathbb{C}$ let $G(X, D)$ be the graph on X in which two points are adjacent when their distance lies in D. A set of real algebraic numbers is countable, and for $d \in F$ the points of E at distance d from 0 are exactly those of dK.

Fractional chromatic numbers. For a graph G with countably many vertices, $\chi_{\mathrm{f}}(G)$ is the least t such that some Borel probability measure on the closed set of independent sets of G in $\{0,1\}^{V(G)}$ gives every vertex probability at least $1/t$ of lying in the random set. For finite G this is the usual fractional chromatic number, and by weak-$*$ compactness $\chi_{\mathrm{f}}(G)$ is the supremum over the finite subgraphs of G. We write $\chi_{\mathrm{f},0}(\mathbb{R}^2, D)$ for the supremum of $\chi_{\mathrm{f}}(H)$ over finite graphs H drawn in the plane with all edge lengths in D, the *finitary* fractional chromatic number of [15] when $D = \{1\}$, and $\chi_{\mathrm{f}}(\mathbb{R}^2, D)$ for the infimum of $\sum_I w(I)$ over finitely supported weightings $w \geq 0$ of independent sets of $G(\mathbb{C}, D)$ with $\sum_{I \ni x} w(I) \geq 1$ for every point x. Clearly $\chi_{\mathrm{f},0} \leq \chi_{\mathrm{f}}$. Since F is real closed, Tarski's transfer principle lets every finite graph drawn with edge lengths in $D \subseteq F$ be redrawn with vertices in E; so, for $D \subseteq F$,

$$\chi_{\mathrm{f},0}(\mathbb{R}^2, D) = \chi_{\mathrm{f}}(G(E, D)) \quad \text{and} \quad \chi(\mathbb{R}^2, D) = \chi(G(E, D)).$$

Densities. The upper density of a measurable $A \subseteq \mathbb{C}$ is

$$\overline{\delta}(A) = \limsup_{R \to \infty} \frac{|A \cap B(0,R)|}{|B(0,R)|}.$$

A set *avoids* D if no two of its points are at a distance in D, and $m_D(\mathbb{R}^2)$ is the supremum of $\overline{\delta}(A)$ over measurable sets A avoiding D. For finite D it is also the supremum of the densities of periodic sets avoiding D (Lemma 10; the same proof works with squares in place of discs), so it does not matter whether densities are taken over discs, as here, or over squares, as in [17, 2]. A measurable set A *weakly avoids* D if

$$\int_{B(0,R)} \int_{S^1} 1_A(x)\, 1_A(x+du)\, d\sigma(u)\, dx = 0 \qquad \text{for all } d \in D,\ R > 0,$$

where σ is the normalized arc length. A *weak measurable k-colouring avoiding D* is a measurable map $c : \mathbb{C} \to \{1, \dots, k\}$ whose classes weakly avoid D.

3. The transfers

From [18] we use Lemma 4.1 (the continuous vectors of a measure-preserving action of E form L^2 of a factor, and the translations extend to a strongly continuous action T_x, $x \in \mathbb{R}^2$, on it), Theorem 2.3 (every K-invariant Borel probability measure on the character group of the discrete group E that gives measure zero to the continuous characters is Haar measure), the measurable sampling step in the proof of Proposition 4.2, and Lemma 4.3. Lemma 4.1 and Theorem 2.3 involve no distance; Lemma 4.3 and the sampling step concern the unit circle, and we redo them at radius d (Lemma 7 and the proofs below).

Lemma 7. *If a measurable set A weakly avoids D, then the set A^* of points of density one of A avoids D, and $|A \setminus A^*| = 0$.*

Proof. This is [18, Lemma 4.3] at radius d. Let $x,\ y = x + v \in A^*$ with $|v| = d \in D$. For small $h > 0$ choose x' uniformly in $B(x, h)$ and, independently, $v' = du'$ with u' uniform on the arc $\{u \in S^1 : |du - v| < h\}$. Then $P(x' \notin A) \to 0$. The point $x' + v'$ is a mixture of uniform points of discs of radius h, so it has a density at most $1/(\pi h^2)$, and it lies in $B(y, 2h)$; hence $P(x' + v' \notin A) \le |B(y, 2h) \setminus A|/(\pi h^2) \to 0$. For small h both points lie in A with positive probability, while the law of (x', u') is absolutely continuous with respect to $dx\, d\sigma(u)$. This contradicts weak avoidance. The second statement is Lebesgue's density theorem. □

Theorem 8. *Let $D \subseteq F$ be a set of positive real algebraic numbers and $k \ge 1$. If $G(E, D)$ has a proper k-colouring, the plane has a weak measurable k-colouring avoiding D; conversely, such a colouring gives a proper k-colouring of $G(\mathbb{C}, D)$. In particular $\chi(\mathbb{R}^2, D) \ge 1/m_D(\mathbb{R}^2)$.*

Proof. This is the proof of [18, Thm. 1.3] with the unit circle replaced by the circles of radii $d \in D$.

(1) Let $\Omega_D \subseteq \{1, \dots, k\}^E$ be the closed set of proper colourings of $G(E, D)$, nonempty by hypothesis. Translations by E and rotations by K preserve it. The group $E \rtimes K$ is amenable, so averages of a point mass along a Følner sequence have an invariant weak-$*$ limit μ.
(2) Let f_i be the indicator that the origin has colour i. Properness gives $\langle f_i, T_z f_i \rangle = 0$ for $z \in dK$, $d \in D$.
(3) By [18, Lemma 4.1], $f_i = p_i + b_i$ where p_i is the conditional expectation on the factor of continuous vectors; $0 \le p_i \le 1$, $\sum_i p_i = 1$, both parts are rotation-fixed, and the spectral measure of b_i is K-invariant and null on the continuous characters. By [18, Thm. 2.3] it is a multiple of Haar measure, so $\langle b_i, T_z b_i \rangle = 0$ for $z \in E \setminus \{0\}$; the cross terms vanish because the continuous subspace and its orthogonal complement are translation invariant. Hence $\langle p_i, T_z p_i \rangle = 0$ for $z \in dK$.
(4) The T_x act strongly continuously and dK is dense in the circle of radius d, so $\langle p_i, T_x p_i \rangle = 0$ whenever $|x| \in D$.
(5) The sampling step of [18, Prop. 4.2] gives jointly measurable fields $H_i(x, \omega)$ representing $T_x p_i$. Since D is countable, a single sample ω makes all the integrals

$$\int_{B(0,m)} \int_{S^1} H_i(x, \omega) H_i(x + du, \omega)\, d\sigma(u)\, dx \qquad (m \in \mathbb{N},\ d \in D,\ \text{all } i)$$

vanish and has $\sum_i H_i(x,\omega) = 1$ for almost every x. Giving x the least i with $H_i(x,\omega) > 0$ defines the weak measurable colouring.

(6) Conversely, by Lemma 7 the density-one points of the classes form disjoint sets avoiding D whose union is conull. Every finite configuration has a translate inside that union, so every finite subgraph of $G(\mathbb{C}, D)$ is k-colourable, and the de Bruijn–Erdős theorem [7] gives a proper colouring. One of the k sets has upper density at least $1/k$, so $k \geq 1/m_D(\mathbb{R}^2)$. □

Theorem 9. *For every set $D \subseteq F$ of positive real algebraic numbers, $\chi_{\mathrm{f},0}(\mathbb{R}^2, D) \geq 1/m_D(\mathbb{R}^2)$.*

Proof. Let $t \geq \chi_{\mathrm{f},0}(\mathbb{R}^2, D) = \chi_{\mathrm{f}}(G(E, D))$ and let P be a Borel probability measure on the compact metrizable space Ω of independent sets of $G(E, D)$ with $P(z \in I) \geq 1/t$ for all $z \in E$. Averaging the push-forwards of P under $E \rtimes K$ along a Følner sequence and taking a weak-$*$ limit gives an invariant probability μ with $\mu(0 \in I) \geq 1/t$, since each push-forward has this property and the event is clopen. Let $f = 1[0 \in I]$. It is rotation-fixed, and $\langle f, T_z f\rangle = \mu(0 \in I,\ z \in I) = 0$ for $z \in dK$, $d \in D$, because I is independent. Steps 3 and 4 of the proof of Theorem 8 apply to the single function f, with $\sum_i p_i = 1$ (which is used only in Step 5) replaced by $\mathbb{E}_\mu p = \mathbb{E}_\mu f$: for $p = \mathbb{E}(f \mid \text{continuous factor})$ we get $0 \leq p \leq 1$, $\mathbb{E}_\mu p \geq 1/t$ and $\langle p, T_x p\rangle = 0$ whenever $|x| \in D$.

Take a jointly measurable $H : \mathbb{C} \times \Omega \to [0,1]$ with $H(x,\cdot) = T_x p$ in $L^2(\mu)$ for almost every x (the sampling step of [18, Prop. 4.2]). For $m \in \mathbb{N}$ and $d \in D$,

$$\int_\Omega \int_{B(0,m)} \int_{S^1} H(x,\omega) H(x+du,\omega)\, d\sigma(u)\, dx\, d\mu(\omega) = \int_{B(0,m)} \int_{S^1} \langle T_x p, T_{x+du} p\rangle\, d\sigma(u)\, dx = 0,$$

so, D being countable, for μ-almost every ω all these inner integrals vanish. Let $X_n(\omega) = |B(0,n)|^{-1} \int_{B(0,n)} H(x,\omega)\, dx \in [0,1]$. Since $\mathbb{E}_\mu H(x,\cdot) = \langle T_x p, 1\rangle = \langle p, 1\rangle \geq 1/t$ for almost every x ($T_x 1 = 1$ and T_x is unitary), $\mathbb{E}_\mu X_n \geq 1/t$, and the reverse Fatou lemma gives $\mathbb{E}_\mu \limsup_n X_n \geq 1/t$. Hence $\limsup_n X_n \geq 1/t$ on a set of positive measure, and we choose ω in it and in the conull set above. Let $A = \{x : H(x,\omega) > 0\}$. Then $1_A \geq H(\cdot,\omega)$, so $\overline{\delta}(A) \geq 1/t$; and $H(x,\omega)H(x+du,\omega) = 0$ for almost every (x,u) gives $1_A(x) 1_A(x+du) = 0$ almost everywhere, so A weakly avoids D. By Lemma 7, A^* avoids D and $\overline{\delta}(A^*) \geq 1/t$. Hence $m_D(\mathbb{R}^2) \geq 1/t$. □

4. The upper bound and the corollaries

Lemma 10. *Let D be finite, $M = \max D$ and $\varepsilon > 0$. There are $\eta > 0$ and a periodic set A' with period lattice $N\mathbb{Z}^2$, each period a finite union of squares, that avoids $D_\eta = D + [-\eta, \eta]$ and has density at least $m_D(\mathbb{R}^2) - \varepsilon$.*

Proof. Let A be measurable, avoiding D, with $\overline{\delta}(A) > m_D - \varepsilon/4$, and fix $L > 0$. For the radii R realising the upper density, with R large, the squares of the grid $L\mathbb{Z}^2$ meeting $B(0,R)$ (at most $\pi(R + \sqrt{2}L)^2/L^2$ of them) have average density of A above $m_D - \varepsilon/2$, so one of them, S, has $|A \cap S| \geq (m_D - \varepsilon/2)L^2$. By inner regularity there is a compact $C \subseteq A \cap S$ with $|C| \geq (m_D - 3\varepsilon/4)L^2$. Its distance set $\{|x - y| : x, y \in C\}$ is compact and misses the finite set D, so it misses $D_{2\eta}$ for some $\eta \in (0,1)$. Let U be the union of the closed squares of a grid of squares of diameter less than $\eta/2$ that meet C. It lies in the $\eta/2$-neighbourhood of C, so it avoids D_η, and $|U| \geq |C|$. The set $A' = U + N\mathbb{Z}^2$ with $N = L + M + 2$ avoids D_η, since points of different copies are more than $M + 1$ apart, and has density $|U|/N^2 \geq m_D - \varepsilon$ once L is large. □

Proof of Theorem 1. By Theorem 9, $\chi_{\mathrm{f}}(\mathbb{R}^2, D) \geq \chi_{\mathrm{f},0}(\mathbb{R}^2, D) \geq 1/m_D$, and $\chi \geq \chi_{\mathrm{f},0}$ always. Let A' be as in Lemma 10, with density $\delta \geq m_D - \varepsilon$ and fundamental square $Q = [0, N)^2$. For $n \geq 1$ take the n^2 translates $A' + s$, $s \in (N/n)\mathbb{Z}^2 \cap Q$; each avoids D. A point x lies in $A' + s$ if and only if $x - s \in A'$, so the number of these translates containing x is the number of points of the grid $x - (N/n)\mathbb{Z}^2$ in one period of A'. One period is a finite union of squares, so its boundary meets $O(n)$ cells of that grid, uniformly in x, and the count is at least $\delta n^2 - O(n)$ for every x. Giving each translate the weight $1/(\delta n^2 - O(n))$ is a fractional colouring of $G(\mathbb{C}, D)$ with total weight tending to $1/\delta$. Hence $\chi_{\mathrm{f}}(\mathbb{R}^2, D) \leq 1/(m_D - \varepsilon)$ for every $\varepsilon > 0$. □

Proof of Corollaries 2–5. Corollary 2 is Theorem 1 with $D = \{1\}$, together with $0.229364 \leq m_1 \leq 0.2470$ [5, 1]; [1, §8] proves $m_1 \leq (w_T + \nu)/(1 - \mu)$ with $w_T = 0.24697262945998308$, $\nu = 10^{-5}$ and $\mu = 6 \cdot 10^{-5}$, which is less than 0.2469975.

Corollary 3. If X is a set of n points, its unit-distance graph G has $\chi_{\mathrm{f}}(G) \le \chi_{\mathrm{f},0}(\mathbb{R}^2) = 1/m_1(\mathbb{R}^2)$, so $\alpha(G) \ge n/\chi_{\mathrm{f}}(G) \ge m_1(\mathbb{R}^2)\, n$. Conversely, [15, Thm. 2] states $\chi_{\mathrm{f},0}(\mathbb{R}^2) = 1/\alpha_1(\mathbb{R}^2)$, where $\alpha_1(\mathbb{R}^2)$ is the infimum of $\alpha(G)/|V(G)|$ over the unit-distance graphs G of finite point sets in the plane, so $\inf_n f(n)/n = \alpha_1(\mathbb{R}^2) = m_1(\mathbb{R}^2)$ by Corollary 2. Placing two point sets far apart shows $f(a+b) \le f(a) + f(b)$, so by Fekete's lemma $f(n)/n$ converges to $\inf_n f(n)/n = m_1(\mathbb{R}^2)$.

Corollary 4. The finite bounds follow from Theorem 8 and the density bounds of Table 1. For the last statement, Bukh's Theorem 1 [2] gives $m_{\{1,d\}}(\mathbb{R}^2) \to m_1(\mathbb{R}^2)^2 \le 0.2470^2 < 1/16$ as $d \to \infty$; Theorem 8 then gives $\chi(\mathbb{R}^2, \{1, d\}) \ge 17$ for large algebraic d. Corollary 6 follows in the same way from Table 2.

Corollary 5. Bukh's Theorem 1 gives $m_D(\mathbb{R}^2) \to m_1(\mathbb{R}^2)^k$ for $D = \{1, t, \dots, t^{k-1}\}$ as $t \to \infty$. For integers t, Theorem 8 gives $\chi(\mathbb{R}^2, D) \ge 1/m_D(\mathbb{R}^2)$; since $\chi(\mathbb{R}^2, D)$ is an integer and $1/m_D(\mathbb{R}^2) \to m_1(\mathbb{R}^2)^{-k}$, we get $\chi(\mathbb{R}^2, D) \ge m_1(\mathbb{R}^2)^{-k}$ for large t. Restricting a colouring of $\mathbb{R}^3$ to a plane gives $\chi(\mathbb{R}^3, D) \ge \chi(\mathbb{R}^2, D)$. For the supremum, Theorem 1 gives $\chi_{\mathrm{f}}(\mathbb{R}^2, D) = 1/m_D(\mathbb{R}^2)$ for finite $D \subseteq F$; Bukh proves $m_D(\mathbb{R}^2) \ge m_1(\mathbb{R}^2)^k$ whenever $|D| = k$ [2], and the sets $\{1, t, \dots, t^{k-1}\}$ show that the bound $m_1(\mathbb{R}^2)^{-k}$ is approached. (By [17, Thm. 5.1], $m_D(\mathbb{R}^2) \le 2^{-k}$ as soon as consecutive ratios exceed some $r(k)$ that their proof makes effective, which gives $\chi(\mathbb{R}^2, D) \ge 2^k$ for such D made of algebraic numbers.) □

5. Linear-programming certificates

Proposition 11 (Oliveira Filho–Vallentin [17], with clique inequalities)**.** *Let $D = \{d_1, \dots, d_\ell\}$ and let $W_1, \dots, W_r$ be finite planar point sets, each with all pairwise distances in D, and $p_1, \dots, p_r$ points. Suppose $z_0, \dots, z_\ell \in \mathbb{R}$ and $y_1, \dots, y_r \ge 0$ satisfy*

$$g(t) = z_0 + \sum_k z_k J_0(d_k t) + \sum_c y_c \sum_{w \in W_c} J_0(|w - p_c|\, t) \;\ge\; 0 \qquad (t \ge 0),$$

with $s = z_0 + \sum_k z_k + \sum_c y_c |W_c| > 0$. Then $m_D(\mathbb{R}^2) \le (z_0 + \sum_c y_c)/s$.

Proof. By Lemma 10 it suffices to consider a periodic measurable set A of density $\delta > 0$ avoiding D. Its autocorrelation $\varphi(v) = \delta(A \cap (A - v))$ is positive definite, with spectral measure ν on the dual lattice, $\nu(\{0\}) = \delta^2$, total mass δ, so that $\varphi(v) = \int e^{2\pi i \langle \xi, v \rangle}\, d\nu(\xi)$; its radial average is $f(r) = \int J_0(2\pi r|\xi|)\, d\nu(\xi)$, with $f(0) = \delta$ and $f(d_k) = 0$. For a set W with pairwise distances in D and any p, the sets $\{x \in A : x + w - p \in A\}$, $w \in W$, are disjoint subsets of A, so $\sum_w \varphi(w - p) \le \delta$; averaging over rotations of W about p gives $\sum_w f(|w - p|) \le \delta$. Integrating $g(2\pi|\xi|) \ge 0$ against ν, $\delta^2 g(0) \le \int g(2\pi|\xi|)\, d\nu = z_0\delta + \sum_c y_c \sum_w f(|w - p_c|) \le (z_0 + \sum_c y_c)\delta$, and $g(0) = s$. Without clique terms this is [17, Thm. 1.1]; the clique terms are in the spirit of [17, §3.1], which uses simplices, while we also use a rhombus. □

We solve the linear program in floating point, round the coefficients to rationals and verify $g \ge 0$ rigorously. Three programs check every certificate.

- `certify_lp2.py` uses Arb ball arithmetic, with distances and radii enclosed from their exact expressions. On $[0, T]$ it walks with adaptive steps: at a point t it encloses $g(t)$ and moves on by about $g(t)/L$, where $L = \sum |\text{coefficient}| \cdot \text{radius}/\sqrt{2}$ bounds $|g'|$ because $|J_1(x)| \le 1/\sqrt{2}$ for real x [9, 10.14.1]. On $[T, \infty)$ it uses $|J_0(x)| \le \sqrt{2/(\pi x)}$ for $x > 0$: $u = \sqrt{x} J_0(x)$ solves $u'' + (1 + 1/(4x^2))u = 0$, so $u^2 + u'^2/(1 + 1/(4x^2))$ is nondecreasing (Sonin–Pólya) with limit $2/\pi$. The clique distances are checked exactly with a computer algebra system, and the bound is an exact fraction.
- `check2.py`, written separately, repeats the check with 60-digit floating point and its own sweep.
- `check3.py`, adapted from the checker of one of the referees, encloses g by Arb over whole intervals of t, bisecting until each enclosure is nonnegative, and builds the cliques and their radii in exact arithmetic of $\mathbb{Q}(\sqrt{3})$.

For $\{1, \sqrt{3}\}$ Oliveira Filho and Vallentin [17, §4] found the bound 0.170213 without clique terms, and 0.141577 for $\{1, j_{1,2}/j_{1,1}\}$, where $j_{1,k}$ are zeros of J_1; that ratio is presumably transcendental, so Theorem 8 does not apply to it directly, but nearby rationals such as 11/6 and 46/25 do.

D	clique sets	certified bound for $m_D(\mathbb{R}^2)$	sweep	consequence
$\{1,2\}$	$6+2$ triangles	$0.159222 < 1/6$	$[0,1297]$	$\chi \geq 7$
$\{1,\sqrt{3}\}$	triangle, rhombus	$0.162866 < 1/6$	$[0,757]$	$\chi \geq 7$
$\{1,(\sqrt{6}+\sqrt{2})/2\}$	none	$0.166018 < 1/6$	$[0,107]$	$\chi \geq 7$
$\{1,10\}$	none	$0.149599 < 1/6$	$[0,36]$	$\chi \geq 7$
$\{1,11/6\}$	none	$0.140995 < 1/7$	$[0,171]$	$\chi \geq 8$
$\{1,46/25\}$	none	$0.139185 < 1/7$	$[0,175]$	$\chi \geq 8$
$\{1,40\}$	none	$0.115511 < 1/8$	$[0,36]$	$\chi \geq 9$
$\{1,100\}$	none	$0.103302 < 1/9$	$[0,37]$	$\chi \geq 10$

TABLE 1. Certified density bounds (rounded up). For $\{1,2\}$ the clique sets are six unit equilateral triangles and two equilateral triangles of side 2, each at one position; for $\{1,\sqrt{3}\}$, one unit equilateral triangle and one rhombus made of two such triangles (diagonals 1 and $\sqrt{3}$). All three programs accept every certificate.

D	certified bound for $m_D(\mathbb{R}^2)$	sweep	consequence
$\{1,40,1600\}$	$0.046389 < 1/21$	$[0,36]$	$\chi \geq 22$
$\{1,10,100,1000\}$	$0.039641 < 1/25$	$[0,37]$	$\chi \geq 26$
$\{1,100,10^4\}$	$0.037168 < 1/26$	$[0,37]$	$\chi \geq 27$

TABLE 2. Certified density bounds for three and four distances (rounded up), without clique terms. `certify_lp2.py` and `check3.py` accept every certificate; the sweeps have up to $6 \cdot 10^6$ steps.

6. Remarks

(1) The hard input is OpenAI's rigidity theorem [18, Thm. 2.3], which OpenAI formalized in Lean. Our contribution is the observation that their transfer holds for every set of algebraic distances and for fractional colourings, and the consequences of combining it with density bounds.

(2) Davies [6] proves, by a different method, that in $\mathbb{R}^d$ with $d \geq 4$, and for finite D whose elements have rational squares, the chromatic number equals the measurable chromatic number and the independence ratio equals the measurable one; his certificates come from the completely positive framework of DeCorte, Oliveira Filho and Vallentin and are transferred through lattice embeddings. With Bukh's theorem this gives exponential growth in k for $d \geq 4$, and he hopes to extend the method to $d = 2, 3$. Our results concern $d = 2$ (and $d = 3$ for Corollary 5), allow arbitrary algebraic distances, and rest on OpenAI's transfer.

(3) The distances must be algebraic because [18, Thm. 2.3] is a statement about the algebraic plane. This is a limitation of the method, but probably also of the statement: if Bukh's conjecture above holds, the inequality $\chi(\mathbb{R}^2, D) \geq 1/m_D(\mathbb{R}^2)$ fails for some algebraically independent D.

(4) No explicit graph is given. The graph of Dúcz and Varga can in principle be made explicit but is astronomically large; an explicit unit-distance graph of reasonable size with fractional chromatic number above 4, or an explicit graph with edges of lengths 1 and 2 and no proper 6-colouring, remains to be found.

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